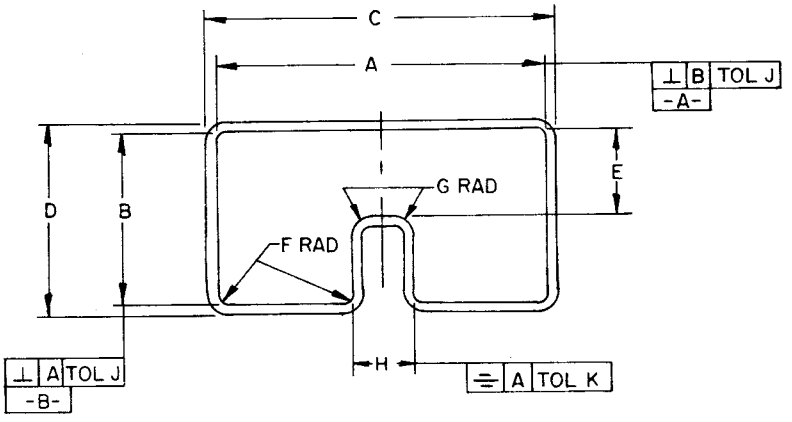
MILITARY SPECIFICATION SHEET

WAVEGUIDES, SINGLE RIDGE

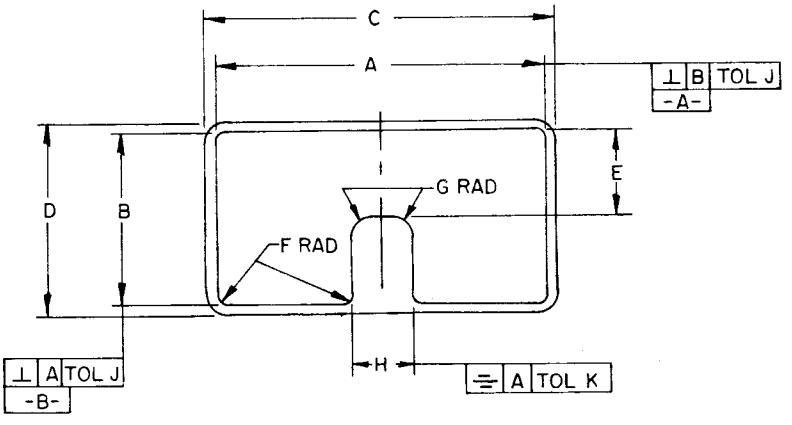
(BANDWIDTH RATIO 3.6:1)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguides described herein shall consist of this document and the latest issue of Specification MIL-W-23351.



Configuration 1.



Configuration 2.

FIGURE 1. Waveguide.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, Department of the Navy, ATTN: ELEX 5043, Washington, D.C. 20360 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5985

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TABLE I. Part number dimensions and material.

Part number M23351/1-	Dimensions 1/			
	A	B	C	D
001 2/ 002 003 004	31.218 ±.020 (792.94) (.51)	14.048 ±.020 (356.82) (.51)	31.468 Max (799.29)	14.298 Max (363.17)
005 006 007 008	12.542 ±.010 (318.57) (.25)	5.644 ±.010 (143.36) (.25)	12.792 Max (324.92)	5.894 Max (149.71)
009 010 011 012	8.677 ±.010 (220.40) (.25)	3.905 ±.010 (99.19) (.25)	8.927 ±.015 (226.75) (.38)	4.155 ±.015 (105.54) (.38)
013 014 015 016	3.494 ±.004 (88.75) (.10)	1.572 ±.004 (39.93) (.10)	3.654 ±.009 (92.81) (.23)	1.732 ±.009 (43.99) (.23)
017 018 019 020 2/	2.422 ±.004 (61.52) (.10)	1.090 ±.004 (27.69) (.10)	2.582 ±.009 (65.58) (.23)	1.250 ±.009 (31.75) (.23)
021 3/ 022 023 024	.968 ±.003 (24.59) (.08)	.436 ±.003 (11.07) (.08)	1.068 ±.008 (27.13) (.20)	.536 ±.008 (13.61) (.20)
025 026 027 028	.6780 ±.0025 (17.221) (.064)	.3050 ±.0025 (7.747) (.064)	.7780 ±.0075 (19.761) (.191)	.4050 ±.0075 (10.287) (.191)
029 030 031 032 3/	.273 ±.001 (6.93) (.03)	.123 ±.001 (3.12) (.03)	.353 ±.006 (8.97) (.15)	.203 ±.006 (5.16) (.15)

See footnotes at end of table.

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TABLE I. Part number dimensions and material - Continued.

Part number M23351/1-	Dimensions (continued) 1/						
	E	F Max	G ±10%	H	J	K	Material
001 2/ 002 003 004	2.402 ±.004 (61.01) (.10)	.065 (1.65)	.480 (12.19)	5.307 ±.010 (134.80) (.25)	.020 (.51)	.010 (.25)	Aluminum alloy Copper alloy Copper Silver alloy
005 006 007 008	.965 ±.003 (24.51) (.08)	.053 (1.35)	.193 (4.90)	2.132 ±.007 (54.15) (.18)	.010 (.25)	.007 (.18)	Aluminum alloy Copper alloy Copper Silver alloy
009 010 011 012	.6680 ±.0025 (16.967) (.064)	.047 (1.19)	.134 (3.40)	1.475 ±.005 (37.47) (.13)	.010 (.25)	.005 (.13)	Aluminum alloy Copper alloy Copper Silver alloy
013 014 015 016	.2690 ±.0015 (6.833) (.038)	.047 (1.19)	.054 (1.37)	.5940 ±.0025 (15.088) (.064)	.004 (.10)	.0025 (.063)	Aluminum alloy Copper alloy Copper Silver alloy
017 018 019 020 2/	.186 ±.001 (4.72) (.03)	.047 (1.19)	.037 (.94)	.412 ±.002 (10.46) (.05)	.004 (.10)	.002 (.05)	Aluminum alloy Copper alloy Copper Silver alloy
021 3/ 022 023 024	.7500 ±.0005 (19.050) (.013)	.031 (.79)	.015 (.38)	.165 ±.001 (4.19) (.03)	.003 (.08)	.001 (.03)	Aluminum alloy Copper alloy Copper Silver alloy
025 026 027 028	.5200 ±.0005 (13.208) (.013)	.015 (.38)	.010 (.25)	.115 ±.001 (2.92) (.03)	.0025 (.063)	.001 (.03)	Aluminum alloy Copper alloy Copper Silver alloy
029 030 031 032 3/	.0210 ±.0005 (.533) (.013)	.015 (.38)	.004 (.10)	.0460 ±.0005 (1.168) (.013)	.001 (.03)	.0005 (.013)	Aluminum alloy Copper alloy Copper Silver alloy

1/ Dimensions are in inches. Metric equivalents (in parentheses) are given for general information only and are based upon 1.00 inch = 25.4 mm.
2/ The waveguide shall be in accordance with configuration 1.
3/ The waveguide shall be in accordance with configuration 2.

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REQUIREMENTS:

Design, construction, and physical dimensions: See figure 1 and table I.
Material: See table I.
Part number: M23351/1- (dash number from table I).

NOTES:

Engineering information and frequency range: See table II.
Cross reference: See table III.

TABLE II. Engineering information.

Part number M23351/1-	Suggested frequency range TE_{10} mode	Cutoff frequency		Theoretical attenuation at $f = \sqrt{3}\, f_{c_{10}}$ 1/ 2/	Theoretical power handling capacity at $f = \sqrt{3}\, f_{c_{10}}$ 3/
		TE_{10} mode	TE_{20} mode		
	GHz	GHz	GHz	dB/ft	kW
001-XXX	0.108- 0.390	0.092	0.404	0.0024	14,500
002-XXX				0.0024	
003-XXX				0.0016	
004-XXX				0.0017	
005-XXX	0.270- 0.970	0.229	1.006	0.0098	2,348
006-XXX				0.0096	
007-XXX				0.0065	
008-XXX				0.0070	
009-XXX	0.390- 1.40	0.331	1.454	0.0168	1,124
010-XXX				0.0166	
011-XXX				0.0112	
012-XXX				0.0120	
013-XXX	0.970- 3.50	0.822	3.611	0.0658	182.2
014-XXX				0.0650	
015-XXX				0.0438	
016-XXX				0.0469	
017-XXX	1.40 - 5.00	1.186	5.210	0.114	87.56
018-XXX				0.113	
019-XXX				0.0758	
020-XXX				0.0812	
021-XXX	3.50 -12.40	2.966	13.030	0.451	13.99
022-XXX				0.445	
023-XXX				0.300	
024-XXX				0.321	
025-XXX	5.00 -18.00	4.237	18.613	0.771	6.857
026-XXX				0.761	
027-XXX				0.513	
028-XXX				0.549	
029-XXX	12.40 -40.00	10.508	46.162	3.019	1.115
030-XXX				2.981	
031-XXX				2.008	
032-XXX				2.150	

1/ $f_{c_{10}}$ = cutoff frequency for TE_{10} mode.

2/ Actual attenuation of waveguides may be considerably greater depending upon operating frequency and temperature. For further information or to calculate for frequencies other than those tabulated, see S. Hopfer, "The Design of Ridged Waveguides", IRE Transactions on Microwave Theory and Techniques, MTT-3 pp 20-29, October 1955, and MIL-HDBK-216, "Transmission Lines and Fittings," chapter 6. Typical resistivities of materials (at 20° C): aluminum alloy (6061) = 4.0 microhms-cm; copper alloy (brass) = 3.9 microhm-cm; copper - 1.77 microhm-cm; silver alloy (coin silver) = 2.03 microhm-cm.

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3/ Actual power handling capability of waveguides may be considerably less depending upon waveguide temperature, altitude, operating frequency, etc. For further information or to calculate values for frequencies other than those tabulated, see S. Hopfer, "The Design of Ridged Waveguides", IRE Transactions on Microwave Theory and Techniques, MTT-3 pp 20-29, October 1955, and MIL-HDBK-216, "Transmission Lines and Fittings", chapter 6. The tabulated values were calculated based upon the E field produced breakdown in a nonpressurized air dielectric waveguide under continuous wave (CW) conditions. The breakdown strength of air was considered to be 15000 volts per centimeter, corner radii considered.

TABLE III. Cross reference (type designation to part number).

Type designation	Part number M23351/1-	Type designation	Part number M23351/1-
WRS108U36A	001	WRS140D36A	017
WRS108U36B	002	WRS140D36B	018
WRS108U36C	003	WRS140D36C	019
WRS108U36S	004	WRS140D36S	020
WRS270U36A	005	WRS350D36A	021
WRS270U36B	006	WRS350D36B	022
WRS270U36C	007	WRS350D36C	023
WRS270U36S	008	WRS350D36S	024
WRS390U36A	009	WRS500D36A	025
WRS390U36B	010	WRS500D36B	026
WRS390U36C	011	WRS500D36C	027
WRS390U36S	012	WRS500D36S	028
WRS970U36A	013	WRS124C36A	029
WRS970U36B	014	WRS124C36B	030
WRS970U36C	015	WRS124C36C	031
WRS970U36S	016	WRS124C36S	032

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army -
Navy - OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR
Navy - MC, AS
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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